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MILITARY SPECIFICATION

# VALVE, EXHALATION, OXYGEN MASK, PRESSURE COMPENSATING, WITH SAFETY RELIEF

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers one type of pressure compensating oxygen mask exhalation valve with safety relief.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| BB-A-1034 | Air, Compressed, For Breathing Purposes |
| QQ-A-200/3B | Aluminum Alloy Bar, Rod, Shapes, and Tube, Extruded, 2024 |
| QQ-B-626 | Brass, Leaded and Non-Leaded, Rod, Shaped, Forgings and Flat Products with Finished Edges (Bar and Strips) |
| WW-T-799 | Tube, Copper, Seamless (For Use with Solder-Type or Flared-tube Fittings) |

FSC 1660

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SPECIFICATIONS

Federal (Continued)

| | |
|---|---|
| ZZ-R-765 | Rubber, Silicone, Low-and-High Temperature and Tear Resistant |
| PPP-B-636 | Box, Fiberboard |
| PPP-T-45 | Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-I-6699 | Intercommunication Set, AN-AIC-10, Bench, Pre-Flight and Flight Test of |
| MIL-B-7883 | Brazing of Steels, Copper, Copper Alloys, and Nickel Alloys, Aluminum and Aluminum Alloys |
| MIL-A-8625 | Anodic Coatings, for Aluminum and Aluminum Alloys |
| MIL-M-9434 | Microphone, Dynamic M-32B-AIC |
| MIL-R-19121 | Regulators, Aviator's Miniature Oxygen Breathing |
| MIL-F-22191 | Films, Transparent, Flexible, Heat Sealable, for Packaging Applications |
| MIL-B-22205 | Bag, Transparent, Flexible, Heat Sealable for Packaging Applications |
| MIL-O-27210 | Oxygen, Aviator's Breathing, Liquid and Gas |
| MIL-R-81553 | Regulator, Chest Mounted, 100 Percent Oxygen, Positive Pressure |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-9 | Screw Thread Conventions and Methods of Specifying |
| MIL-STD-100 | Engineering Drawing Practices |



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STANDARDS

Military (Continued)

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-130 | Identification Marking of U. S. Military Property |
| MS 22001 | Mask Assemblies, Oxygen, Pressure Breathing |
| MS 22056 | Valve, Check, Oxygen Mask, Inlet |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials Standard

| | |
|---|---|
| D 1149 | Method of Test for Accelerated Ozone Cracking of Vulcanized Rubber |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

American Standards Association (ASA) Standard

| | |
|---|---|
| USAS Y14.5-1966 | Dimensioning and Tolerancing for Engineering |
| USAS B46.1-1962 | Surface Texture |

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Qualification - The exhalation valve furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3, 4.3.1, 6.3, and 6.3.1).



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In addition, the retention of the qualification for the exhalation valve on the applicable qualified products list shall be dependent on periodic verification of continued compliance with the requirements of this specification (see 4.3 and 4.3.2).

3.2 First article - Unless otherwise specified, the exhalation valve furnished under this specification shall be a product which has been inspected and has passed the first article inspection specified in 4.4 through 4.4.2.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended. Any material that is adversely affected by continued use with oxygen shall not be used.

3.3.1 Age - Elastomer components, except silicone, shall not be more than 12 months old from the cure date of manufacture to the date of delivery to any Government Service or to any airframe or accessory manufacturer (see 6.2(g)).

3.3.2 Elastomer components - The elastomer components shall be composed of an ozone-resistant composition, which shall not bloom, and shall meet the specified ozone-resistance performance requirements.

3.3.3 Protective treatment - When materials are used in the construction of the valves that are subject to deterioration when exposed to environmental conditions which are likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coating which might crack, chip, or scale during normal service life or under extremes of environmental conditions shall not be used.

3.4 Design and construction - The design and construction of the exhalation valve shall be in accordance with Figure 1. The valve shall be designed to control the pressure within the oxygen mask assembly and to respond automatically to pressure changes in the oxygen breathing regulators. The valve shall be designed to snap into place in the expiratory chamber of the mask assembly and shall form a leak-tight fit. Suction in the breathing cavity section of the mask shall not cause the valve to open. There shall be no undue flutter or vibration of the valve under service conditions.

3.4.1 Safety relief - The exhalation valve shall be designed to relieve pressure in excess of 18 inches of water.

3.5 Performance -

3.5.1 Dumping - The valve, when tested as specified in 4.7.2, shall relieve any pressure greater than 16 inches of water and shall not allow the pressure on the inlet side of the valve to exceed 18 inches of water.



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3.5.2 Leakage, suction - The valve, when tested as specified in 4.7.3, shall not permit leakage in excess of 30 cubic centimeters per minute.

3.5.3 Resistance to flow - The valve, when tested as specified in 4.7.4, shall not exceed the maximum limits shown in the characteristic curve on Figure 2.

3.5.4 Burst pressure - The diaphragm of the valve, when tested as specified in 4.7.5, shall not rupture at less than 5 psig.

3.5.5 High temperature operation - The valve, after being conditioned as specified in 4.7.6, shall operate satisfactorily and shall not show any evidence of mechanical or material failure.

3.5.6 Low temperature operation - The valve, after being conditioned as specified in 4.7.7, shall operate satisfactorily and shall not show any evidence of mechanical or material failure.

3.5.7 Extreme low temperature exposure - The valve, when tested as specified in 4.7.8, shall operate satisfactorily and shall not show any evidence of mechanical or material failure.

3.5.8 Vibration - The valve, when tested as specified in 4.7.9, shall operate satisfactorily and not show any evidence of mechanical or material failure.

3.5.9 Endurance - The valve, when tested as specified in 4.7.10, shall operate satisfactorily and shall not show any evidence of mechanical or material failure.

3.5.10 Decompression - The valve, when tested as specified in 4.7.11, shall not allow facial cavity pressure to exceed 2.0 psig within the first 100 milliseconds and shall cause the facial cavity pressure to drop to the normal pressure breathing schedule of the regulator within a period from 300 milliseconds to 10 seconds.

3.5.11 Effect on communications - The valve, when tested as specified in 4.7.12, during speech periods at the specified altitudes, shall not whistle, chatter, or create noise excessively to a degree which interferes with communication. The valve shall not modulate the speech signal because of valve flutter.

3.5.12 Ozone-resistance test (elastomer components) - The test slabs, when tested as specified in 4.7.13, shall not show any evidence of checking, cracking, or damage.

3.6 Interchangeability - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.



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3.7 Weight - The weight of the valve shall not exceed 1 ounce.

3.8 Finish - Aluminum and aluminum alloy parts shall be anodized in accordance with MIL-A-8625.

3.9 Identification of product - The valves shall be marked for identification in accordance with MIL-STD-130, except that the Federal Stock Number shall be omitted.

3.9.1 Special marking - A nameplate, decalcomania or other suitable material, shall be marked in white letters on black background and attached to the valve face. The following information shall be provided on the nameplate:

VALVE, EXHALATION, OXYGEN MASK, PRESSURE
COMPENSATING, WITH SAFETY RELIEF

3.10 Workmanship - The exhalation valve shall be uniform in quality and shall be free from irregularities, defects, or foreign matter which could adversely affect safety, performance, reliability, or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The examination and testing of the exhalation valve shall be classified as follows:

(a) Qualification inspection - Qualification inspection consists of examinations and tests performed on samples submitted for approval as qualified products (see 3.1, 4.3, 4.3.1, 6.3, and 6.3.1).

(1) Retention of qualification. Retention of qualification consists of periodic verification to determine compliance of the qualified product with the requirements of this specification (see 3.1, 4.3, and 4.3.2).

(b) First article inspection - First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 3.2 and 4.4 through 4.4.2).



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(c) Quality conformance inspection - Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots to the requirements set forth in this specification (see 4.5).

4.3 Qualification -

4.3.1 Inspection - Qualification inspection of the exhalation valve shall consist of all the examinations and tests of this specification.

4.3.1.1 Samples - Qualification samples shall consist of five valves and three elastomer test slabs, 6 by 6 by 0.075 inches, composed of the identical ozone-resistant composition and cure that was used in manufacturing the elastomer components. The qualification samples shall be forwarded to the Supply Officer, Naval Air Development Center, Warminster, Pennsylvania 18974, Attention: Director, Aerospace Crew Equipment Department, ACED, or to a laboratory designated by the ACED (see 6.3). The samples shall be plainly identified by securely attached durable tags marked with the following information:

Samples for qualification inspection
VALVE, EXHALATION, OXYGEN MASK, PRESSURE COMPENSATING, WITH SAFETY RELIEF
Manufacturer's designation or number
Sample submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-V-81552A and number under authorization (reference authorizing letter and number, see 6.3).

4.3.2 Retention - The retention of qualification shall consist of periodic verification to determine compliance of the qualified exhalation valve with the requirements of this specification. The time and method of periodic verification shall be specified by the activity responsible for the Qualified Products List and shall be included in the Notice of Qualification letter.

4.4 First article inspection - The first article inspection of the exhalation valve shall consist of the following examinations and tests:

Visual examination
Dimensions
Dumping
Leakage, suction
Resistance to flow
Explosive decompression
*Effect on communications*

4.4.1 First article samples - Unless otherwise specified, as soon as practicable after the award of the contract or order, the manufacturer shall submit



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four valves. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of these valves from contract to contract, submission of further first article inspection samples on the new contract may be waived at the discretion of the procuring activity (see 6.2(c)). Approval of the first article inspection samples or the waiving of the first article inspection does not preclude the requirements for performing the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2(d)).

4.4.2 Upon completion of the first article inspection, all the applicable inspection reports and, when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the cognizant Government activity. One of the approved first article inspection sample valves will be returned to the manufacturer for use in monitoring production. The other three valves will be destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under the contract.

4.5 Quality conformance inspection - The sampling and inspection levels shall conform to MIL-STD-105. Quality conformance inspection shall consist of the following examinations and tests:

Visual examination
Dimensions
Dumping
Leakage, suction
Resistance to flow
Preparation for delivery

4.5.1 Sampling -

4.5.1.1 Inspection lot -

4.5.1.1.1 Valve - An inspection lot size shall be expressed in units of one valve made essentially under the same conditions and from the same materials and components. The sample unit shall be one valve.

4.5.1.1.2 Preparation for delivery - An inspection lot size shall be expressed in units of one fully prepared shipping container, containing valves fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing valves fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examinations of valves - The sample size, acceptance criteria, tests, and examinations required for the valves shall be as specified in Table I.



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TABLE I

SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS AND EXAMINATIONS OF THE VALVES

| INSPECTION | PARAGRAPH | | SAMPLE SIZE | ACCEPTANCE CRITERIA 1/ |
|---|---|---|---|---|
| | REQUIREMENT | METHOD | | |
| Visual examination (see classification of defects) | 3.5 through 3.5.8, 3.9 and 3.10 | 4.7.1.1 | Every valve for critical defects. Inspection Level II for minor defects. | Reject all units with any critical defects. An acceptable quality level of 2.5 defects per hundred units for minor defects. |
| Dimensions | 3.4 and Figure 1 | 4.7.1.1.1 | Inspection Level S-2 | An acceptable quality level of 1.0 defect per 100 units. |
| Dumping | 3.5.1 | 4.7.2 | Every valve | Reject all defective units. |
| Leakage, suction | 3.5.2 | 4.7.3 | Every valve | Reject all defective units. |
| Resistance to flow | 3.5.3 | 4.7.4 | Every valve | Reject all defective units. |
| Preparation for delivery | Section 5 | 4.7.1.2 | Inspection Level S-2 | An acceptable quality level of 2.5 defects per 100 units. |

1/ The sampling plan acceptance numbers shall apply collectively to all the characteristics within a stated acceptable quality level.

4.6 Test conditions -

4.6.1 Temperature and pressure - Unless otherwise specified, tests shall be conducted at local ambient temperature and barometric pressure. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests. Any test results obtained at conditions other than normal temperature and pressure (NTP) conditions shall be mathematically corrected to NTP conditions. NTP conditions are 70° F and 29.92 inches of mercury. In the case of pressure drop measurements, the flow setting for conducting the test shall be corrected to NTP conditions before the test is conducted.



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4.6.2 Gas - Unless otherwise specified the gas used in testing the valves shall be oxygen conforming to MIL-O-27210, Type I. When specified, air conforming to BB-A-1034 may be used.

4.6.3 Test stand - The following tests shall be conducted by means of a test stand conforming to the specified figure:

| | |
|---|---|
| Dumping | Figure 3 |
| Resistance to flow | Figure 3 |
| Leakage, suction | Figure 4 |
| Endurance (1st 10,000 cycles) | Figure 5 |
| Endurance (2nd 10,000 cycles) | Figure 6 |
| Explosive decompression | Figure 7 |

The other tests shall be conducted by means of any suitable apparatus. Alternate test equipment, capable of performing the identical tests specified herein, may be used when specifically approved by the procuring activity (see 6.2(h)). Unless otherwise specified, tests shall be conducted with the reference axis of the valve in the vertical position.

4.6.3.1 Caution - The following precautions shall be taken:

The systems specified in 4.6.3 shall be free of oil, grease, and other foreign matter.
Make all lines as short as possible.
Systems shall be leaktight at 15 inches of water.
Do not allow flexible hose to kink.
All tubing not otherwise specified shall be 1/4 inch I.D.

4.6.3.2 Pressure requirement - Pressure source A, as shown on Figure 3 and Figure 6, shall be capable of being regulated from zero to 20 inches of water pressure with an inlet air pressure of 50 psig. Vacuum source, as shown in Figure 4, shall be capable of being regulated within the range of zero to 15 inches of water vacuum at a flow of 1 liter per minute.

4.6.4 Test fixture -

4.6.4.1 Fixture number 1 - When fixture number 1 is specified, it shall consist of an assembly of pieces manufactured in accordance with details shown on Figure 8, and assembled as shown on Figure 9, sequence 1. The fixture shall be tightened securely, using rubber gaskets for seals, and installed in the specified test stand.



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4.6.4.2 Fixture number 2 - When fixture number 2 is specified, it shall consist of an assembly of pieces manufactured in accordance with details shown on Figure 8 and assembled as shown on Figure 9, sequence 2. The fixture shall be tightened securely, using rubber gaskets for seals, and installed in the specified test stand.

4.7 Inspection methods -

4.7.1 Visual examinations -

4.7.1.1 Valve - Every valve shall be examined visually (for critical defects) to determine conformance to this specification. The classification of defects, Table II, shall be used to classify the defects found.

TABLE II

CLASSIFICATION OF DEFECTS FOR VISUAL EXAMINATION OF THE VALVES

| CRITICAL | MINOR |
|---|---|
| 1. Material imperfections - foreign matter embedded. | 201. Marking - missing, insufficient, incorrect, illegible, or not permanent. |
| 2. Surface - unclean, rough, misaligned, or containing cracks, nicks, or other flaws. | |
| 3. Any component missing, malformed, fractured, or otherwise damaged. | |
| 4. Any component loose or otherwise not securely retained. | |
| 5. Incorrect assembling or improper positioning of components. | |
| 6. Any functioning part that works with difficulty. | |
| 7. Faulty workmanship or other irregularities. | |

4.7.1.1.1 Dimensions - The valves shall be checked dimensionally to determine conformance to the dimensions specified in Figure 1.



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4.7.1.2 Preparation for delivery - Each of the fully prepared shipping containers, containing valves, selected as a sample unit from the lot, shall be examined for defects listed in Table III to determine that the packaging, packing, and marking conform to this specification.

TABLE III

LIST OF DEFECTS FOR PREPARATION FOR DELIVERY

| ITEM | DEFECTS |
|---|---|
| Exterior and interior markings | Missing, incorrect, incomplete, illegible; of improper size, location, sequence; or method of application; markings not the same on the interior and exterior containers. |
| Packaging and packing materials | Any non-conforming component; any component missing, damaged, or otherwise defective. |
| Workmanship | Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, loose strappings, etc.; bulging or distortion of the containers. |
| Exterior and interior weight or content | Number per container is more or less than required; gross or net weight exceeds the requirements. |

4.7.2 Dumping - Fixture number 1 (see 4.6.4.1) shall be assembled as shown on Figure 9, sequence 1 and installed in test stand as shown on Figure 3. Set regulator A by adjusting control valve 1 so that a constant pressure of 16 inches of water, indicated on manometer 1, shall be applied to the pressure-sensory probe of the valve. Under these conditions, a pressure of 16 inches of water shall be maintained on manometer 2. A flow of gas, 10 liters per minute, shall then be passed through the valve in the normal direction of flow. The flow of gas shall then be increased from 10.0 to 100.0 lpm in increments of 10 lpm. During the test, at 10 lpm and with each increase of flow, the pressure read on manometer 2 shall pass the requirements specified in 3.5.1.

4.7.3 Leakage, suction - Fixture number 2 (see 4.6.4.2) shall be assembled as shown on Figure 9, sequence 2 and installed in test stand as shown on Figure 4. Open control valve 1 slowly and adjust vacuum source, indicated on manometer A, to show a pressure within the range of 0.5 to 12 inches of water. The leakage shall be read on rotometer B and shall pass the requirements specified in 3.5.2.



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4.7.4 Resistance to flow - Fixture number 1 (see 4.6.4.1) shall be assembled as shown on Figure 9, sequence 1 and installed in test stand as shown on Figure 3. Disconnect compensating tube from the pressure-sensory probe of the valve. Close control valve 1 and open control valve 2. Adjust control valve 2 to produce a flow through the valve in the normal direction throughout the range of 0 to 100 lpm. During the test, the pressure resistance to flow shall be read on manometer 2 and shall pass the requirements specified in 3.5.3.

4.7.5 Burst pressure - A pressure source capable of being regulated from zero to 25 pounds per square inch (psi) shall be connected to the pressure-sensory probe of the valve. In the line, between the pressure source and the sensor probe, insert a tee connection and connect a mercury manometer calibrated in psi to the tee connection. Pressure shall be applied to the sensor probe and increased until the diaphragm in the valve ruptures. The burst pressure shall be noted and the valve shall pass the requirement specified in 3.5.4.

4.7.6 High temperature operation - The valve shall be installed in a test chamber and the temperature shall be raised to 160 ±5° F and maintained for 3 hours. After the 3-hour stabilization period and while maintained at this temperature, the valve shall be subjected to and pass the dumping and resistance to flow tests. The valve shall also pass the requirements specified in 3.5.5.

4.7.7 Low temperature operation - The valve shall be installed in a test chamber and the temperature shall be lowered to + 20 ±5° F and maintained for 3 hours. After the 3-hour stabilization period and while at this temperature, the valve shall be subjected to and pass the dumping and resistance to flow tests. The valve shall also pass the requirements specified in 3.5.6.

4.7.8 Extreme low temperature exposure - The valve shall be subjected to a temperature of -65 ±5° F for 3 hours. After the conditioning period, the valve shall be returned to room temperature and shall then be subjected to and pass the dumping, leakage, and resistance to flow tests. The valve shall also pass the requirement specified in 3.5.7.

4.7.9 Vibration - The valve shall be placed on a rubber sheet 4 by 4 by 0.188 inches and secured to a frame suitable for mounting on a vibration machine. The valve shall be vibrated for 3 hours at a frequency from 600 to 1100 cycles per minute and a double amplitude of 0.018 to 0.020 inch. During the vibration period, the valve shall be checked for tightness every hour. The valve shall pass the requirements specified in 3.5.8. After the vibration period, the valve shall be subjected to and pass the dumping, leakage, and flow resistance tests.

4.7.10 Endurance - Fixture number 1 (see 4.6.4.1) shall be assembled as shown on Figure 9, sequence 1 and installed in test stand as shown on Figure 5. The valve shall then be subjected to 10,000 cycles of operation. Each cycle shall consist of flowing gas through the valve in the normal direction at a rate varying from zero to 50 and back to zero liters per minute. The rate of cycling shall be 18



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cycles per minute. The fixture shall then be removed from test stand and installed in test stand shown on Figure 6. Turn air inlet pressure on and set air pressure regulator valve so that a pressure of 14 inches of water, indicated on manometer, shall be applied to the pressure-sensory probe of the valve. The valve shall then be subjected to an additional 10,000 cycles of operation under conditions specified above. After the completion of the 20,000 cycles of operation, the valve shall be subjected to and pass the dumping, leakage, and resistance to flow tests.

4.7.11 Decompression - Fixture number 1 (see 4.6.4.1) shall be assembled as shown on Figure 9, sequence 1 and installed in test stand as shown on Figure 7. The test stand shall then be placed in an altitude chamber divided into two compartments. An altitude of 19,000 feet shall be maintained in the first compartment. A second compartment, connected to the first, shall have a volume at least three times that of the first and shall be maintained at an altitude such that after the explosive decompression a final altitude of 45,000 feet shall be obtained. The entire explosive decompression time from 19,000 feet to 45,000 feet shall equalize within 100 milliseconds. The valve shall pass the requirements specified in 3.5.10.

4.7.12 Effect on communications - The valve shall be installed in an MS22001 pressure-breathing oxygen mask that is equipped with an M-32/AIC microphone conforming to MIL-M-9434. The microphone shall be connected to an AN/AIC-10 intercommunication set conforming to MIL-I-6699. The mask shall then be connected to an oxygen regulator that will provide safety pressure during the test. The effects on communications shall be determined by listening qualitatively to speech communication from at least three persons when using the valve at ground level, and at altitudes of 16,000 feet and 32,000 feet. Under each condition, the valve shall pass the requirements specified in 3.5.11.

4.7.13 Ozone-resistance test (elastomer components) - The three test slabs submitted for qualification shall be tested for ozone-resistance. The test apparatus shall be in accordance with ASTM Method No. D1149. The test slabs shall be elongated 20 percent, placed in an ozone-free atmosphere for 24 hours, then placed in the ozone chamber. The chamber shall be adjusted to a temperature of 100 ±2° F, and to give an exposure of ozone concentration of 120 ±10 parts by volume of ozone per million parts by volume of air. The air-ozone velocity in the chamber shall be at least 2 feet per second. The material shall be exposed to these conditions for 60 minutes. The test slabs shall be examined under 10X magnification and pass the requirements specified in 3.5.12.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging - Packaging shall be Level A or C, as specified (see 6.2).



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5.1.1 Level A - Each valve shall be cleaned in accordance with MIL-P-116, Method C-1 and preserved and packaged in accordance with MIL-P-116, Method 1A 13. No contact preservative is permitted. Each valve shall be heat sealed within a clear snug fitting polyethylene bag of 0.00125 ±20 percent inch thick material conforming to MIL-B-22205, Type III. The continuous line marking requirement of paragraph 3.4.1 of MIL-F-22191 is not mandatory. The closing heat seal seam shall be as close as possible to the edge of the bag. Prior to or during the heat sealing operation, excess air shall be expelled to the extent necessary to permit packaging of the required quantity in the container. The sealed valve shall be suitably cushioned and packaged within a close fitting set up or folding paper box. The valve shall be cushioned in such a manner as to prevent damage during storage.

5.1.2 Intermediate packaging - Twenty-four bagged and cushioned valves, packaged as specified in 5.1.1, shall be packaged within a snug fitting intermediate fiberboard box conforming to PPP-B-636, Type CF, Domestic Class, Variety SW, Grade 125. The valves shall be stacked in two tiers in the box. The weight of the contents in each package shall not exceed 20 pounds. Each container shall be constructed and closed in accordance with the appendix to PPP-B-636. The fiberboard container shall not contain any metal fastenings or stitches.

5.1.3 Level C - The valves shall be packaged to afford the minimum degree of protection necessary to prevent corrosion, deterioration, or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.2 Packing - Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A - Sixteen containers (three hundred and eighty-four valves) packaged as specified in 5.1.2, shall be packed as specified in 5.2.2, except that the fiberboard container shall be Weather-Resistant Class, Variety SW, Grade V3c or V3s. In addition, each container shall be reinforced with flat steel strapping or tape banding in accordance with the appendix to PPP-B-636.

5.2.2 Level B - Sixteen containers (three hundred and eighty-four valves) packaged as specified in 5.1.2, shall be packed flat, four in length, two in width, and two in depth within a snug fitting fiberboard container conforming to PPP-B-636, Type CF or SF, Domestic Class, Variety SW, Grade 275. Gross weight shall not exceed 70 pounds. Each container shall be constructed and closed in accordance with the appendix to PPP-B-636. All the seams, corners, and manufacturer's joints of the box shall be sealed with 3-inch wide tape conforming to PPP-T-45, Type III, Grade A, B, or C. The fiberboard container shall not contain any metal fastenings or stitches.



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5.2.3 Level C - All packaged valves which require over-packing for acceptance by the carrier, shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet the uniform freight classification rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Marking - In addition to any special markings required by the contract or order, all unit, intermediate and shipping containers shall be marked in accordance with MIL-STD-129. The date of cure/manufacture (month and year) and the contract or order number shall be indicated on all unit and intermediate containers.

5.3.1 Precautionary marking - The following precautionary marking shall appear on each unit package:

"CAUTION: DO NOT ALLOW CONTAMINANTS OF ANY KIND TO BE USED ON OR ABOUT THE EXHALATION VALVES"

6. NOTES

6.1 Intended use - The exhalation valve is intended for use in connection with a pressure demand oxygen regulator, or any other regulator which will supply oxygen to the user in such a manner that, at altitudes from sea level up to 50,000 feet, a mask pressure higher than the ambient air pressure is continually maintained.

6.2 Ordering data - Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) Quantity desired.

(c) Whether first article inspection is waived (see 4.4.1).

(d) Name and address of the first article inspection laboratory (see 4.4.1).

(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(f) Whether any special markings are required (see 5.3).

(g) Certificate of compliance for the age of the materials and components (see 3.3).



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(h) Whether alternate test equipment may be used for conducting tests (see 4.6.3).

(i) Items of data required (see 6.4).

6.3 Qualification - With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command, Department of the Navy, Washington, D. C. 20360; however, authorization for qualification of products shall be obtained from the Commanding Officer, Naval Air Development Center, Warminster, Pennsylvania, 18974, Attention: Director, Aerospace Crew Equipment Department. Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Aerospace Crew Equipment Department indicating a date on which the samples can be forwarded and also request an authorization number to accompany the samples.

6.3.1 When requested, the manufacturer shall submit engineering drawings and inspection reports in accordance with SD6, Provisions Governing Qualification.

6.4 Data - For the information of Contractors and Contracting Officers, any of the data specified in applicable documents listed in Section 2 of this specification, or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NAVAIR Form 4200/25 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

Custodians:
Navy - AS
Army - AV

Preparing Activity:
Navy - AS
(Project No. 1660-0218)

Review Activities:
Navy - AS
Army - AV



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NOTICE - Review/user information is current as of date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.



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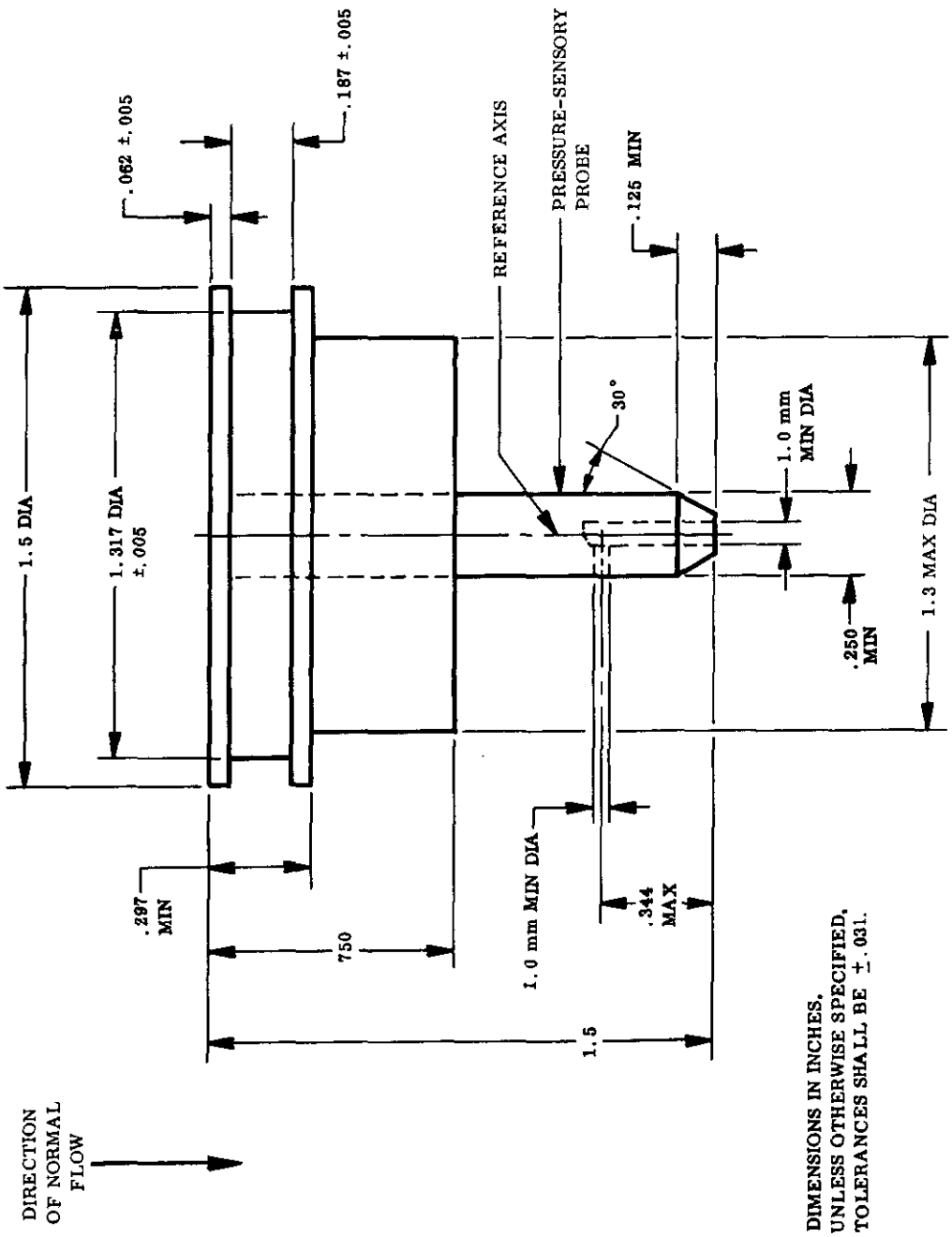



FIGURE 1. VALVE, EXHALATION, OXYGEN MASK, PRESSURE COMPENSATING, WITH SAFETY RELIEF



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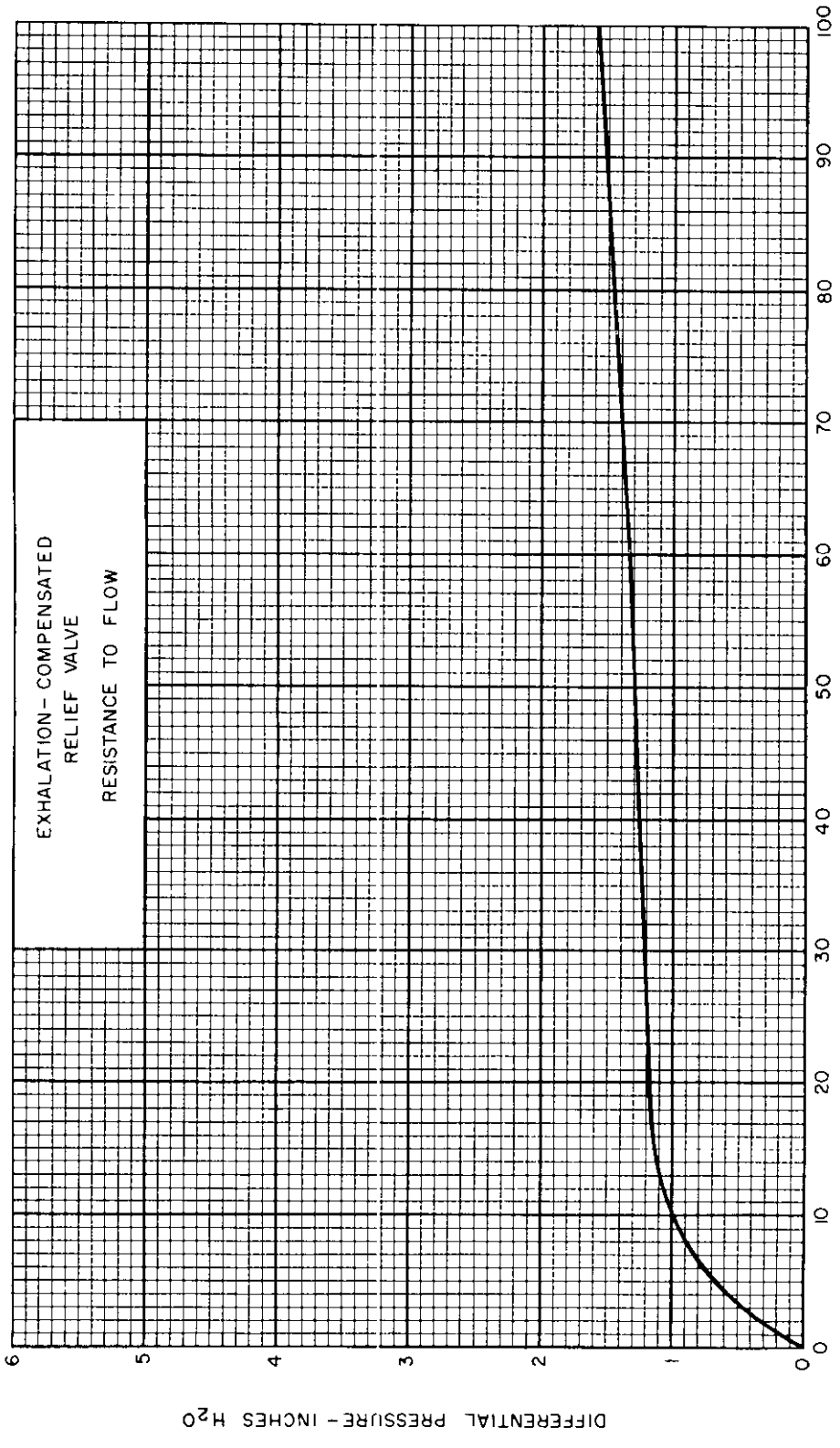



FIGURE 2.



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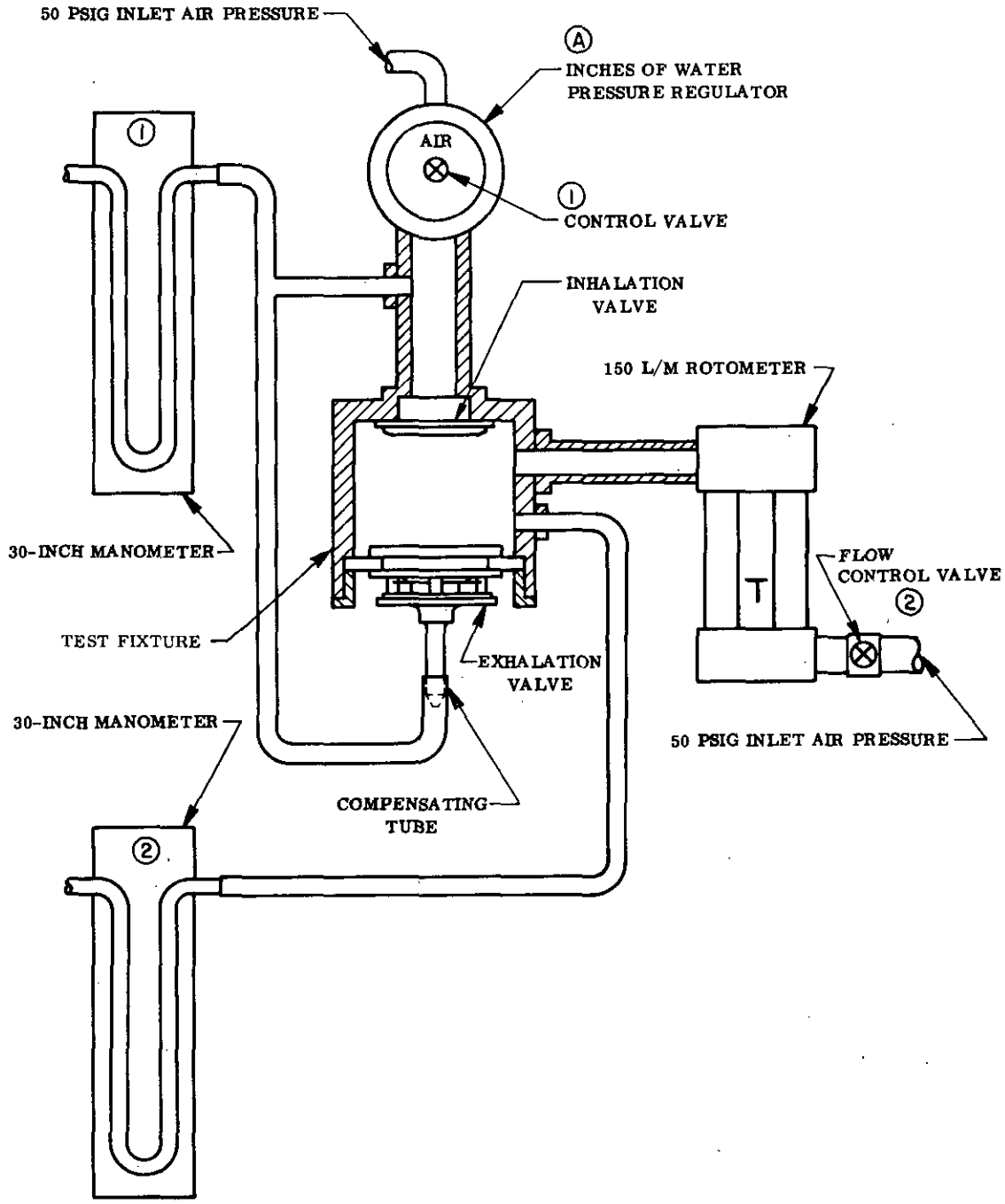



FIGURE 3. TEST STAND - DUMPING, RESISTANCE TO FLOW



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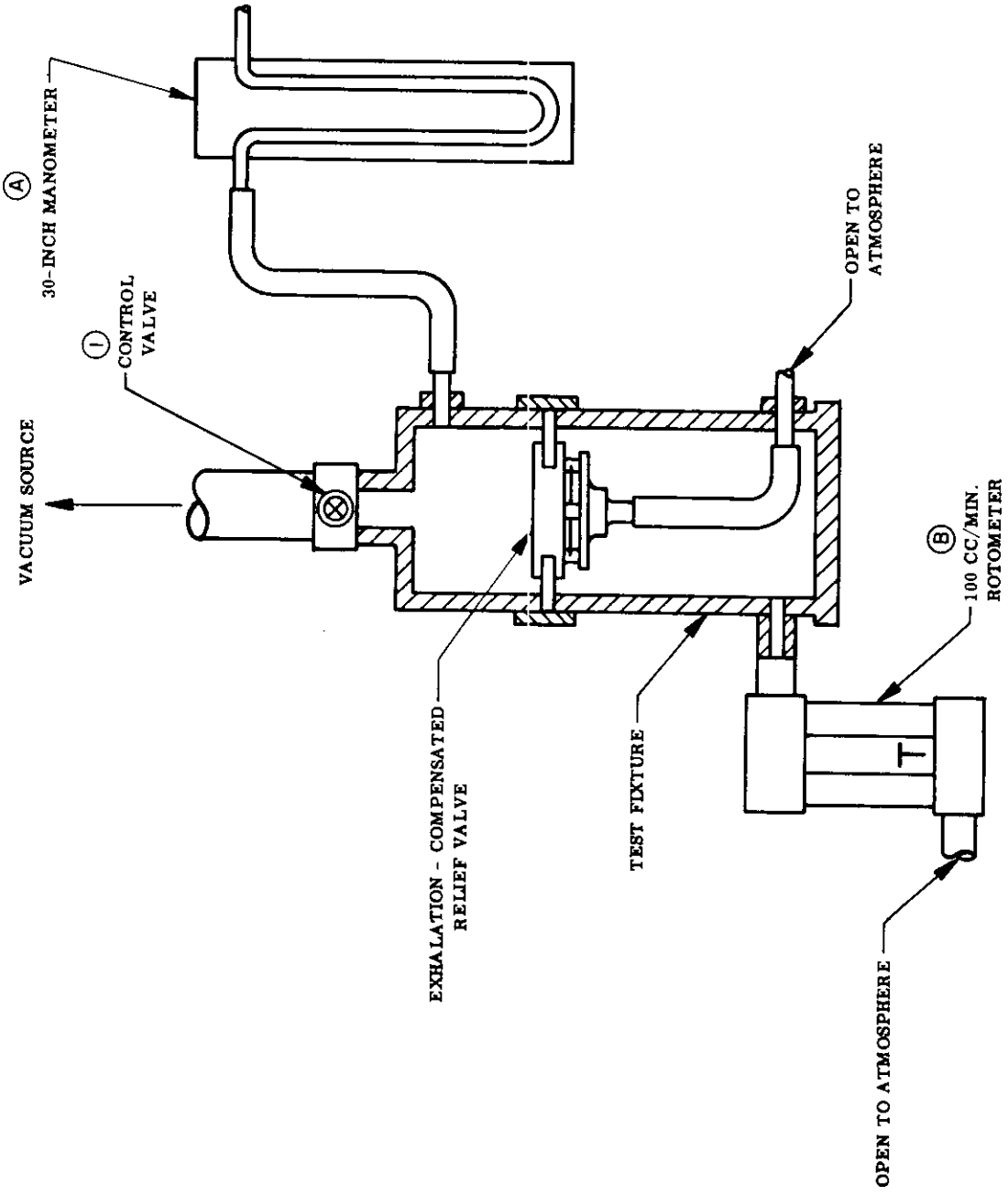



FIGURE 4. TEST STAND - LEAKAGE, SUCTION



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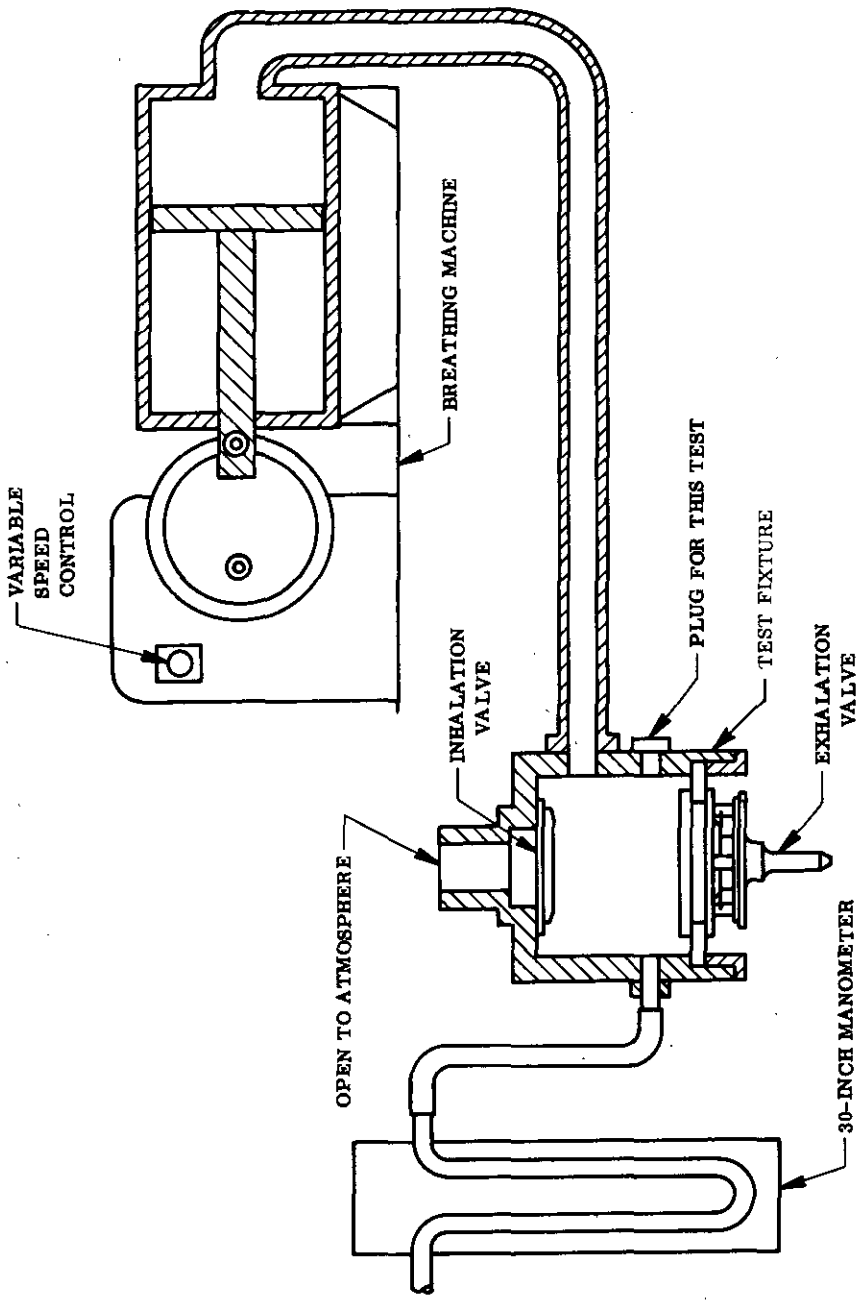



FIGURE 5. TEST STAND - ENDURANCE (1ST 10,000 CYCLES)



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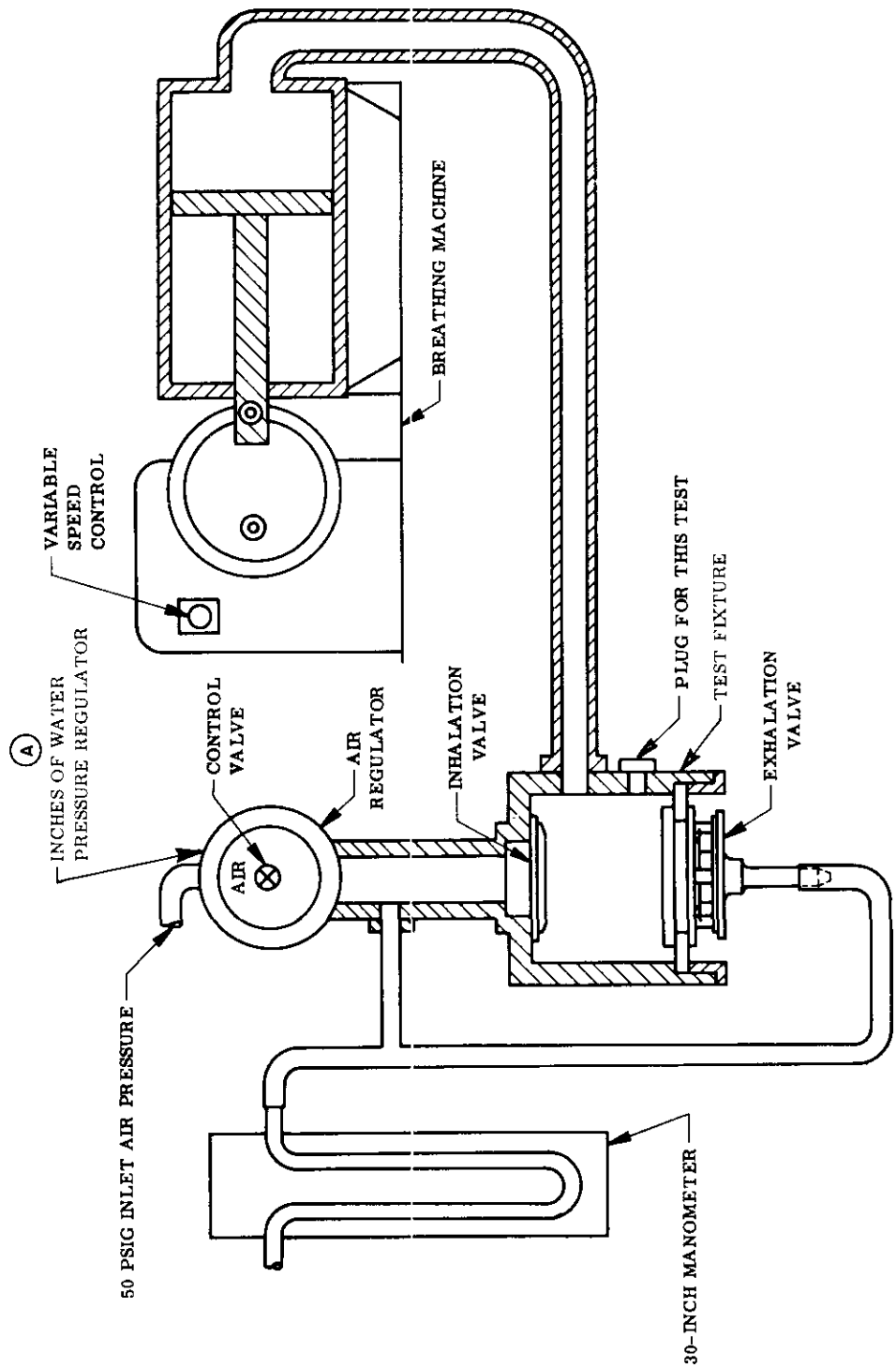



FIGURE 6. TEST STAND - ENDURANCE (2ND 10,000 CYCLES)



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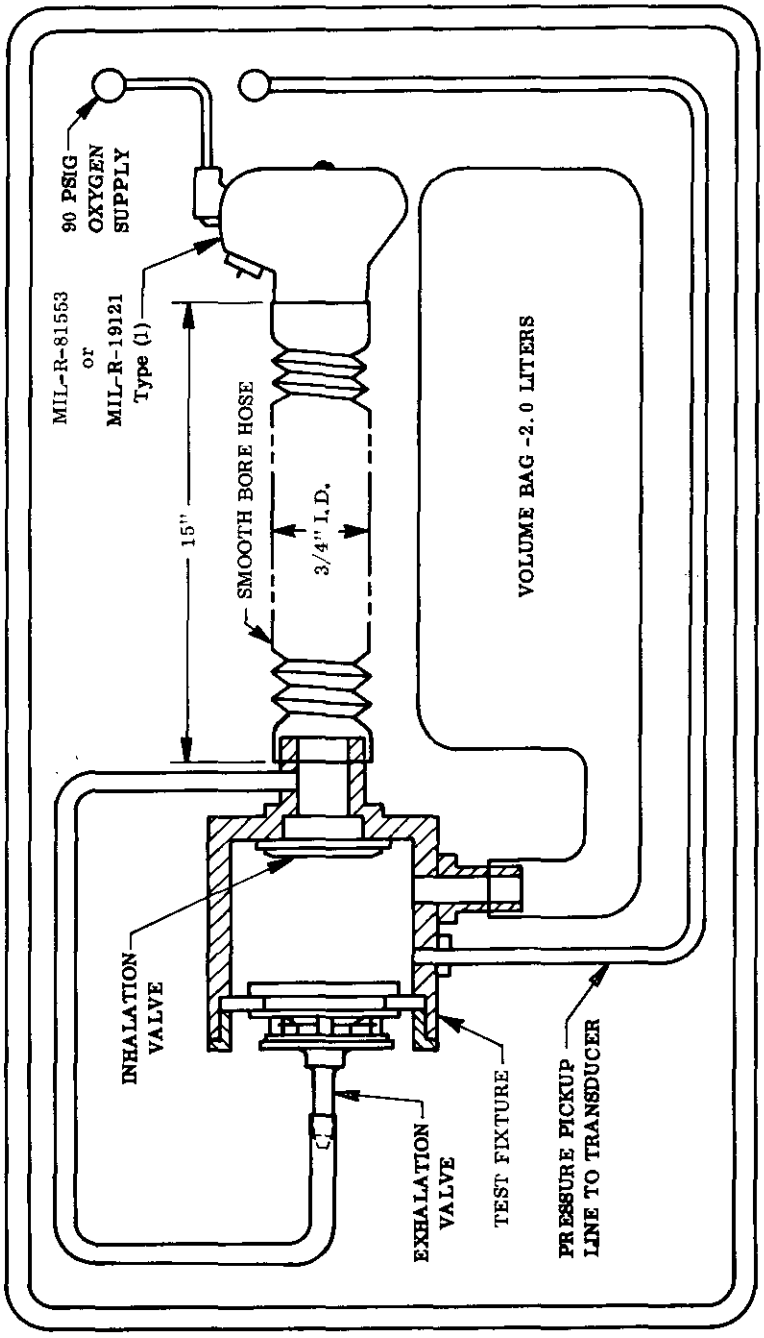
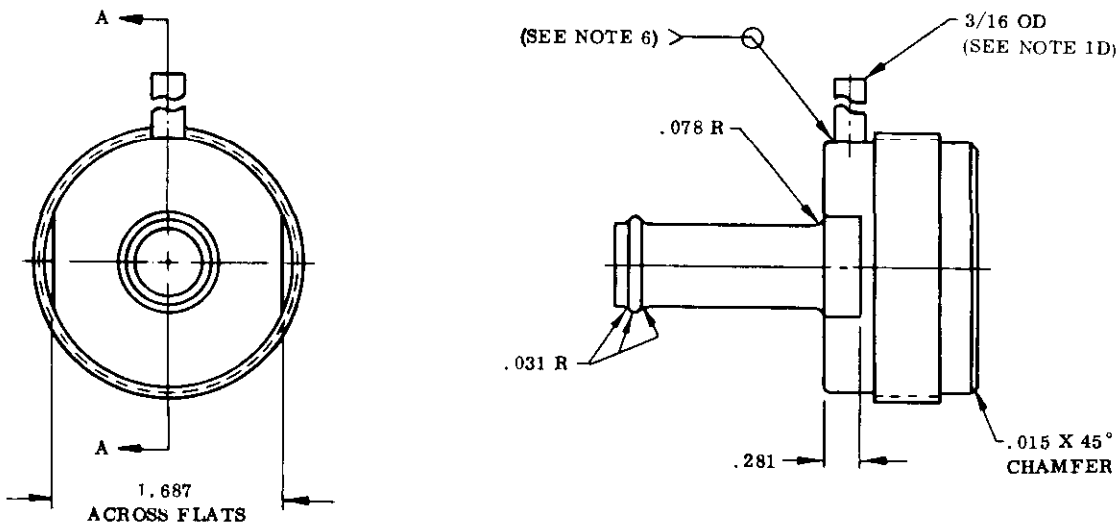


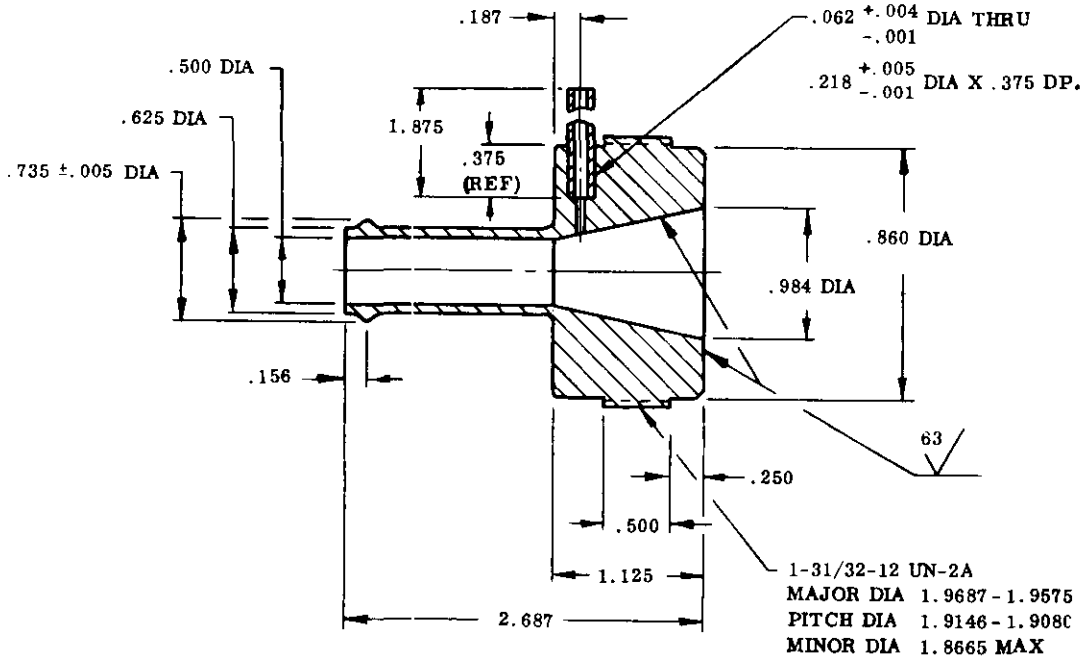

FIGURE 7. TEST STAND - EXPLOSIVE DECOMPRESSION



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INTAKE HOSE FITTING (LEAKAGE TEST)

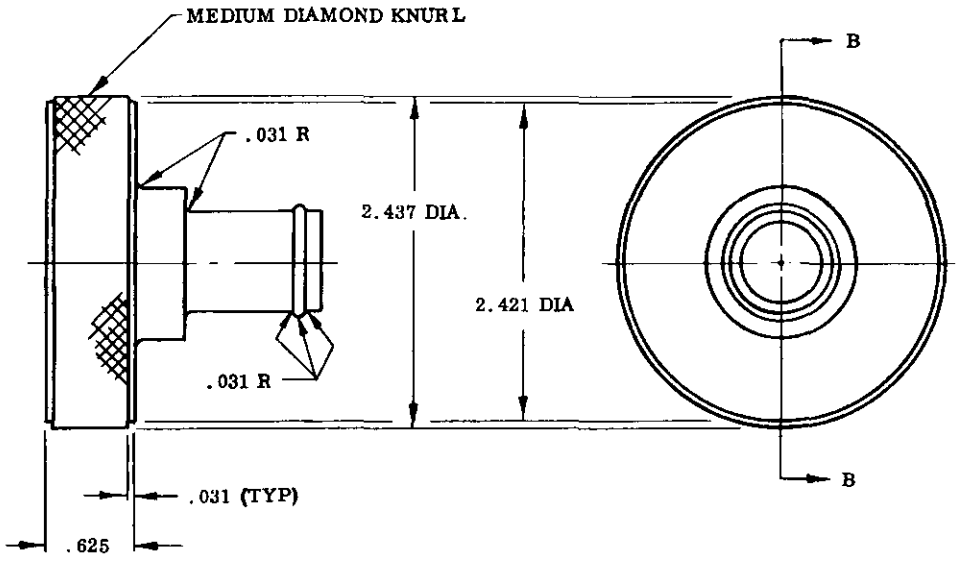



SECTION A-A

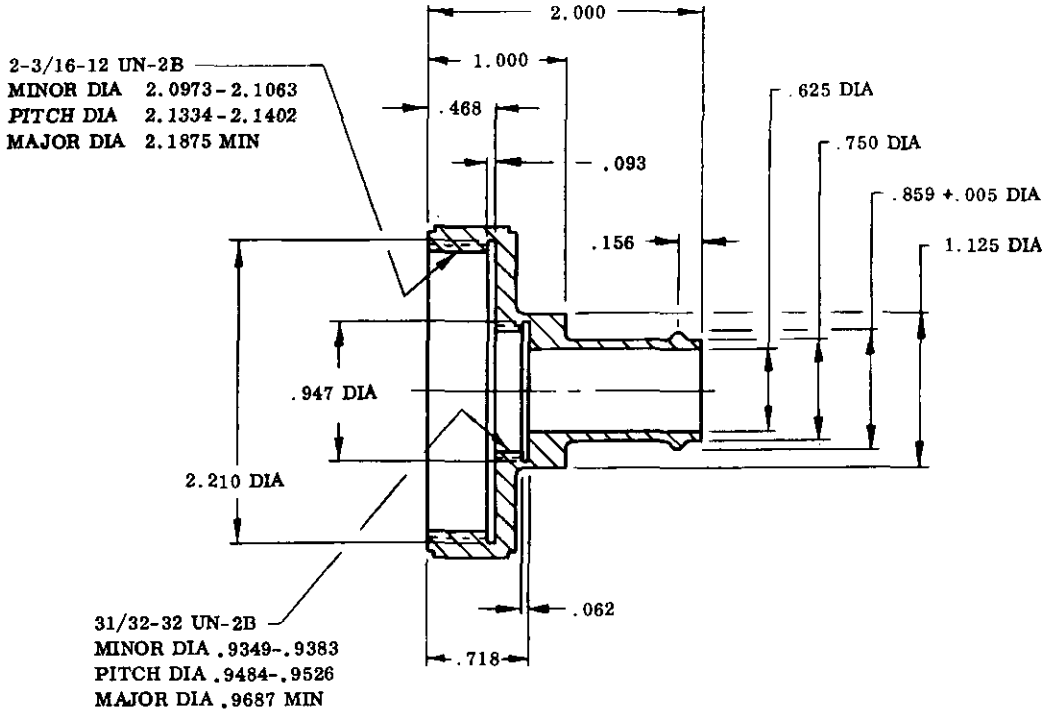
FIGURE 8. TEST FIXTURE DETAILS (SHEET 1)



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INTAKE HOSE FITTING (EXHALATION-INHALATION)




SECTION B-B

FIGURE 8. TEST FIXTURE DETAILS (SHEET 2)



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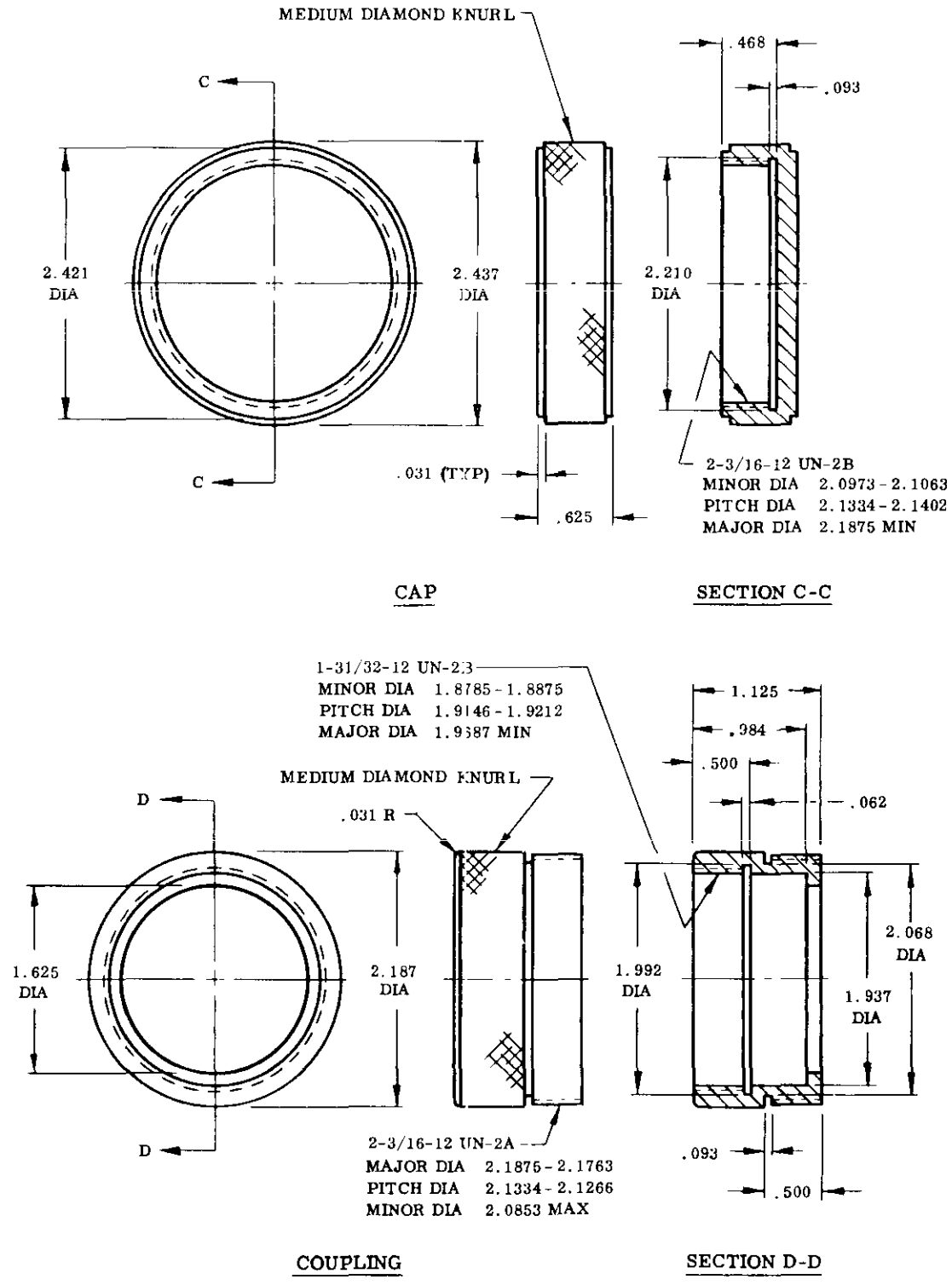
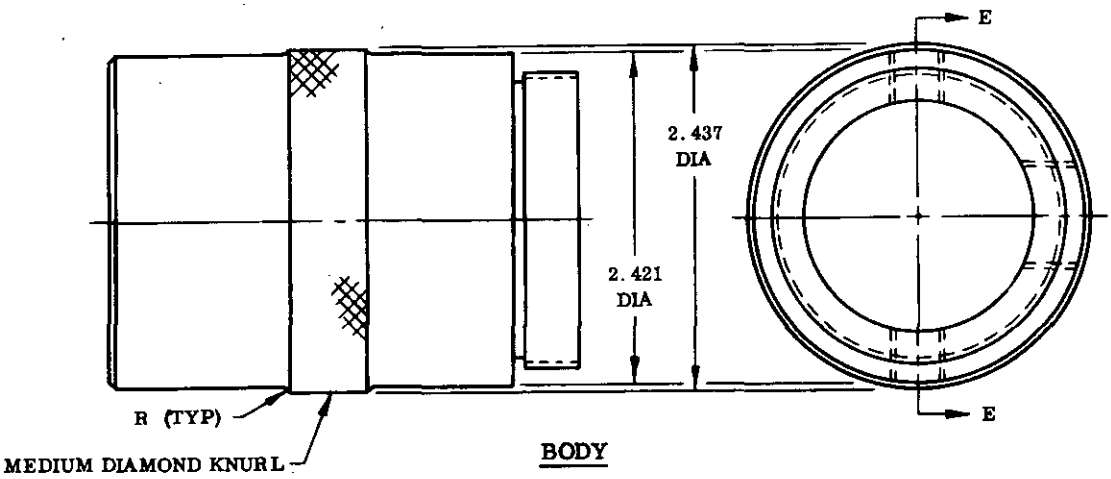


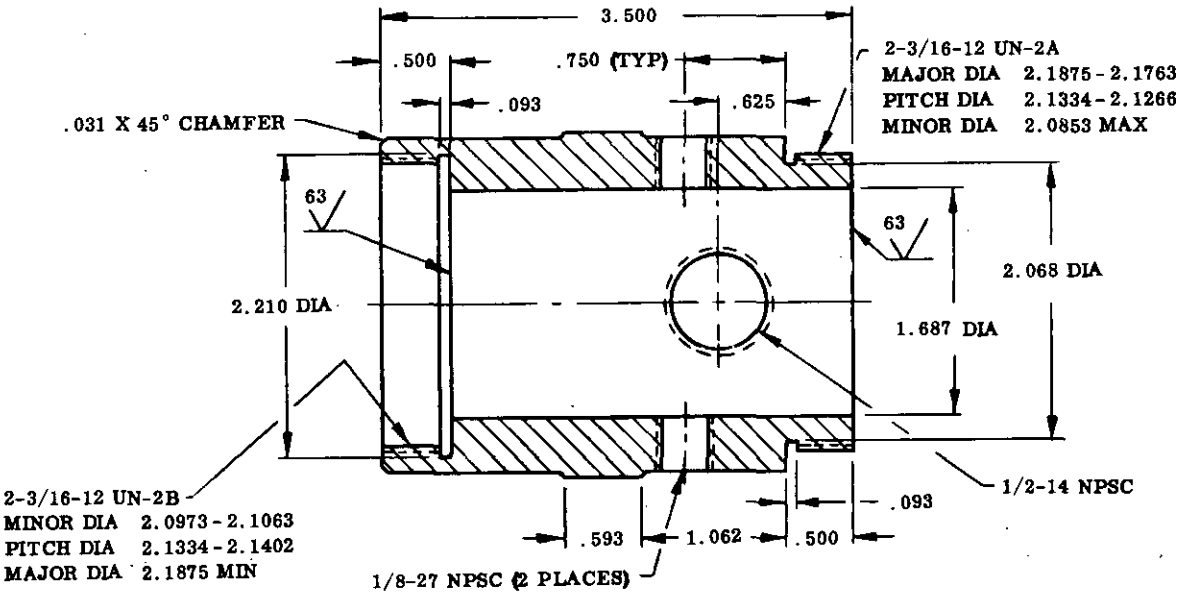

FIGURE 8. TEST FIXTURE DETAILS (SHEET 3)



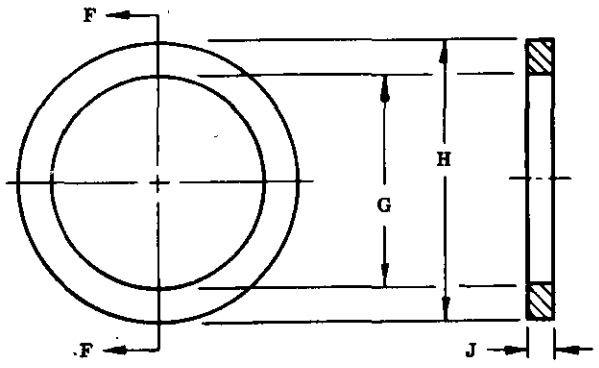
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BODY




SECTION E-E




GASKET

SECTION F-F

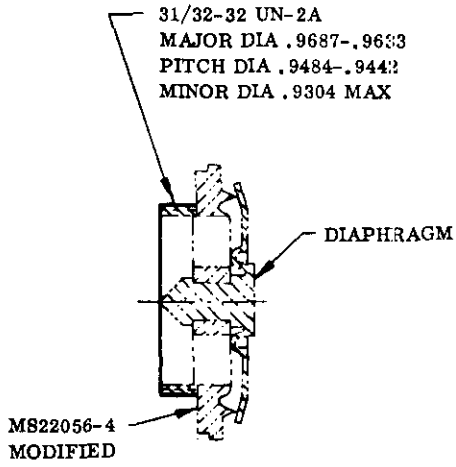
| GASKET TYPES* | | | |
|---|---|---|---|
| DASH NO. | G | H | J |
| -1 | 1.28 | 2.187 | .187 |
| -2 | 1.28 | 1.875 | .187 |
| -3 | 1.625 | 2.187 | .062 |

*SEE NOTE 1C AND FIGURE 9.

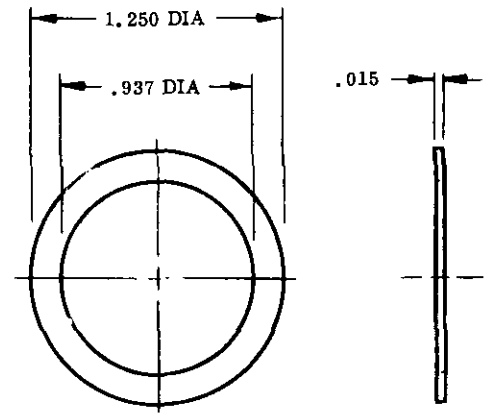
FIGURE 8. TEST FIXTURE DETAILS (SHEET 4)



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INLET CHECK VALVE
(SEE NOTE 7)




INLET CHECK VALVE GASKET

NOTES:

1. MATERIAL: (A) BODY; ALUMINUM ALLOY, 2024 TEMPER TA, QQ-A-200/3B.
   (B) INTAKE HOSE FITTING (LEAKAGE TEST), INTAKE HOSE FITTING (EXHALATION - INHALATION), CAP, COUPLING; BRASS, QQ-B-626, COMP. 622, HALF HARD.
   (C) GASKETS; SILICONE RUBBER, ZZ-R-765.
   (D) SEAMLESS TUBE; COPPER, WW-T-799

2. SURFACE ROUGHNESS: 125 ALL OVER EXCEPT AS NOTED. SURFACE CONDITION FINISH DESIGNATIONS SHALL BE INTERPRETED IN ACCORDANCE WITH USAS B46.1-1962.

3. THREAD DIMENSIONS AND DESIGNATIONS SHALL BE INTERPRETED IN ACCORDANCE WITH MIL-STD-9.

4. REMOVE ALL BURRS AND BREAK ALL SHARP EDGES.

5. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED. TOLERANCES: DECIMALS ±.010, ANGLES ±1° INTERPRET DIMENSIONING AND TOLERANCING IN ACCORDANCE WITH USAS Y14.5-1966.

6. ALL BRAZING SHALL BE IN ACCORDANCE WITH MIL-B-7883.

7. THE INLET CHECK VALVE SHALL BE MS22056-4,WITHOUT CAP AND MODIFIED AS SHOWN.

8. INTERPRET DRAWING IN ACCORDANCE WITH MIL-STD-100.

FIGURE 8. TEST FIXTURE DETAILS (SHEET 5)



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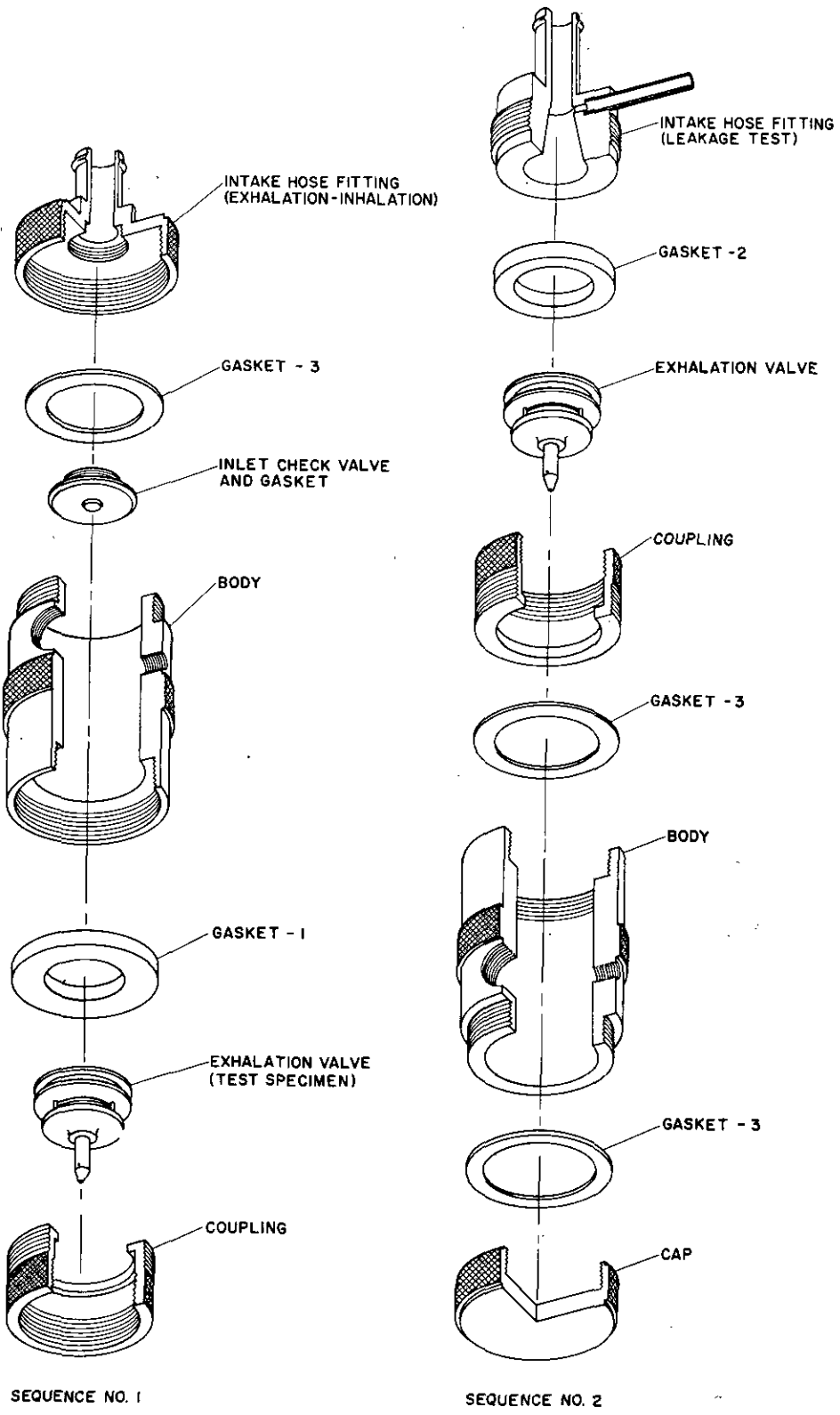



Figure 9. Test Fixture - Method of Assembly



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FOLD

DEPARTMENT OF THE NAVY
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Weapons Engineering Standardization Office (Code X)
Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

FOLD

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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-V-81552A — VALVE, EXHALATION, OXYGEN MASK, PRESSURE COMPENSATING, WITH SAFETY RELIEF

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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